

Bounding the area of surfaces with prescribed boundary

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(joint with Ernst Kuwert)

Background

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Answer. Yes. The unique minimizers are spheres.

$$\frac{1}{2} \int_{\Sigma} \underbrace{|A|^2}_{=\kappa_1^2 + \kappa_2^2} d\mu_g \geq \frac{1}{4} \int_{\Sigma} \underbrace{|H|^2}_{=(\kappa_1 + \kappa_2)^2} d\mu_g \underbrace{\geq}_{\text{Willmore ineq}} 4\pi.$$

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- **Observation.** For a closed surface Σ in $\mathbb{R}^3$, without any restriction on Σ , we **CANNOT** bound its area in terms (*only*) of $\int_{\Sigma} |A|^2 d\mu_g$, namely,

$$|\Sigma| \not\leq C \int_{\Sigma} |A|^2 d\mu_g.$$

Example: A family of 2-spheres $\{S_r(0)\}_{r \in \mathbb{R}_+}$, with

$$\begin{cases} |S_r(0)| = 4\pi r^2 \rightarrow \infty & \text{as } r \rightarrow \infty, \\ \int_{S_r(0)} |A|^2 d\mu_g = 8\pi, & \forall r > 0. \end{cases}$$

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- Existence of minimizers/area bounds of curvature energy (in **compact** Riemannian manifold $(M, \bar{g})$):
 - **Mondino 14'**: **existence of minimizers**: direct method using (unoriented) curvature varifolds (introduced by **Hutchinson 86'**).

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area bounds: *Contradiction argument*. Assuming that the area bound fails. Applying compactness argument to construct a **non-zero curvature varifold** with vanishing generalized second fundamental form.

Assuming Nonexistence of such weak surfaces $\Rightarrow$ **area bounds**
 $\Rightarrow$ **existence of minimizers**.

Background

- Bangert-Kuwert 19': If a **compact** Riemannian manifold $(M^n, \bar{g})$, $n \geq 3$, does not admit totally geodesic surface immersion
 $\Rightarrow$ **area bounds** in terms of $\int_{\Sigma} |A|^2 d\mu_g$.

 $\Rightarrow$ **existence of minimizers** on any **compact** Riemannian manifold.

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Minimization problem with boundary constraint?

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- **Dirichlet boundary condition** (prescribing boundary curve and co-normal):
 - **Novaga-Pozzetta 20'**: **existence of minimizers** of the Willmore energy.
 - Direct method: (unoriented) curvature varifolds with boundary (**Mantegazza 96'**).
 - If the Willmore energy is below the energy threshold 4π .

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 - If the Willmore energy is below the energy threshold 4π .
- **Variational boundary conditions**:
 - **Kuwert-Müller 22'**: free boundary/orthogonal boundary.
 - **Wang-Z.' 25'**: capillary boundary.

Motivation

- **Question.** With a boundary constraint: Given a C^2 -closed curve γ , embedded in $\mathbb{R}^3$.

Can we find a surface (in the weak sense) with boundary given by γ , which minimizes the (in the weak sense) L^2 -curvature energy?

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- **Question.** Given a C^2 -closed curve γ , embedded in $\mathbb{R}^3$. For any immersed surface $F : \Sigma \rightarrow \mathbb{R}^3$ with $F(\partial\Sigma)=\gamma$, can we bound the area of $F(\Sigma)$, in terms (only) of its L^2 -curvature energy $\int_{\Sigma} |A|^2 d\mu_g$?
- Useful for applying the *direct method*.

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 - Useful for applying the *direct method*.
- **Remark.** Prescribing boundary curve but **NOT** prescribing co-normal.

Main results

Theorem (Kuwert-Z'). Given a C^2 -closed **curve** γ , embedded in $\mathbb{R}^3$. If γ is not a plane curve, then for any immersed **surface** $F : \Sigma \rightarrow \mathbb{R}^3$ with $F(\partial\Sigma) = \gamma$, subjects to the following constraint:

$F(\Sigma)$ is contained in a closed set $K \subset \mathbb{R}^3$,

there exists $C > 0$, depending on γ, K , such that

$$|\Sigma| \leq C \int_{\Sigma} |A|^2 d\mu_g.$$

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- For any plane curve, the planar domain delimited by it has 0-curvature energy and positive area;
- The constraint assumption is necessary in this result.

Counterexample: Closed 2-spheres $\{S_r(x_r)\}_{r \in \mathbb{R}_+}$ with radius $r \rightarrow \infty$, containing unit circle $\mathfrak{c}^1$. The larger part separated by $\mathfrak{c}^1$, say $S_r^+(x_r)$, satisfies

$$\begin{cases} |S_r^+(x_r)| = 2\pi r^2 \left(1 + \sqrt{1 - \frac{1}{r^2}}\right) \rightarrow 4\pi r^2, \\ \int_{S_r^+(x_r)} |A|^2 d\mu_g = 4\pi \left(1 + \sqrt{1 - \frac{1}{r^2}}\right) \rightarrow 8\pi. \end{cases}$$

Energy threshold and diameter bound

- Simon's monotonicity formula: for any fixed $x_0 \in \partial\Sigma$,

$$4\pi \leq \frac{1}{4} \int_{\Sigma} |H|^2 d\mu_g + 8 \frac{|\partial\Sigma|}{|x - x_0|}, \quad \forall x \in \Sigma \setminus \partial\Sigma.$$

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Under the **energy threshold constraint** that $\int_{\Sigma} |A|^2 d\mathcal{H}^2 < 8\pi - \delta$ for some $\delta \in (0, 8\pi)$,

$$\frac{1}{16} \int_{\Sigma} \underbrace{|H|^2}_{(\kappa_1 + \kappa_2)^2} d\mu_g \leq \frac{1}{8} \int_{\Sigma} \underbrace{|A|^2}_{\kappa_1^2 + \kappa_2^2} d\mu_g < \pi - \frac{1}{8}\delta,$$

so that

$$\sup_{x \in \Sigma \setminus \partial\Sigma} |x - x_0| \leq 2 \frac{|\partial\Sigma|}{\pi - \frac{1}{16} \int_{\Sigma} |H|^2 d\mu_g}.$$

$\Rightarrow$ **boundedness of support.**

Main results

Theorem (Kuwert-Z'). Given a C^2 -closed curve γ , embedded in $\mathbb{R}^3$.
If

$$\beta := \inf \left\{ \int_{\Sigma} |A|^2 d\mu_g : \Sigma \text{ immersed in } \mathbb{R}^3, \partial\Sigma \subset \gamma \right\} < 8\pi,$$

then β has minimizers.

Sketch of proof: the direct method

Definition (*Weak immersions with curvature and boundary prescribed by γ*): A triple (V, B, Γ) , where

- (weak surfaces) V is a Radon measure on the trivial oriented Grassmannian bundle $\mathbb{R}^3 \times \widetilde{G}(2, 3)$;
- (weak second fundamental form) B is a V -measurable, L^1_{loc} integrable, $\mathbb{R}^3 \times \mathbb{R}^3$ -valued function, defined V -a.e.;
- (weak boundary prescribed by γ): Γ is a Radon measure, supported on a **suitable subspace** of the trivial oriented Grassmannian bundle $\gamma \times \widetilde{G}(2, 3)$.

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When a defining integral formula (*curvature identity w.r.t $\phi \in C^1_c(\mathbb{R}^3 \times \widetilde{G}(2, 3))$*) is satisfied, we call $(V, B, \Gamma) \in \widetilde{CV}_\gamma(\mathbb{R}^3)$ *oriented curvature varifolds with boundary prescribed by γ* .

Sketch of proof: Step 1

Proposition (Compactness): Given $\{(V_j, B_j, \Gamma_j) \in \widetilde{CV}_\gamma(\mathbb{R}^3)\}_{j \in \mathbb{N}}$, if

$$\mathbb{M}(V_j) + \int |B_j|^2 dV_j \leq C, \quad \forall j \in \mathbb{N},$$

then after passing to a subsequence $\{j'\}_{j' \in \mathbb{N}}$, there exists $(V, B, \Gamma) \in \widetilde{CV}_\gamma(\mathbb{R}^3)$, such that

- $V_{j'} \rightarrow V$ as oriented varifolds;
- $\Gamma_{j'} \rightarrow \Gamma$ as oriented varifolds;
- $B \in L^2(V)$, $B_{j'} V_{j'} \rightarrow BV$ as vector-valued Radon measures, with

$$\|B\|_{L^2(V)}^2 \leq \liminf \|B_{j'}\|_{L^2(V_{j'})}^2.$$

Note that we do not assume the **uniform bounds** $\mathbb{M}(\Gamma_j) \leq C$. $\forall j$.

Sketch of proof: Step 2

Lemma (Dichotomy): Let γ be a C^1 -closed, embedded curve in $\mathbb{R}^3$. For any fixed 2-plane $\mathbb{P} \subset \mathbb{R}^3$, if $\gamma \cap \mathbb{P} \neq \emptyset$, then

- either $\gamma \subset \mathbb{P}$;
- or $\mathbb{P} \setminus \gamma$ is an unbounded, connected domain.

Sketch of proof: Step 2

Lemma (Constancy lemma): **If γ is not a plane curve**, then for any $(V, B, \Gamma) \in \widetilde{CV}_\gamma(\mathbb{R}^3)$, the following can not hold simultaneously:

- ❶ $V \neq 0$;
- ❷ *either* $\text{spt}\|V\|$ is bounded, *or* $\mathbb{M}(V) < \infty$;
- ❸ $B = 0$ for V -a.e.

In other words, if (2) (3) hold, then it must be that $V = 0$.

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Proof. By **Lemma (Dichotomy)**, $\mathbb{P} \setminus \gamma$ is an unbounded, connected domain, for any fixed 2-plane $\mathbb{P}$ with $\gamma \cap \mathbb{P}$,

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$$V \underbrace{\Rightarrow}_{\text{disintegration}} \mu_{V, \mathbb{P}} \underbrace{\Rightarrow}_{\text{Constancy theorem}} \mu_{V, \mathbb{P}} \llcorner U = \Theta_U \mathcal{L}^2 \llcorner U, \forall \text{ connected component } U \subset \mathbb{P} \setminus \gamma, \text{ and } \Theta_U \geq 0.$$

□

Sketch of proof

Proposition (non-trivial minimizer): **If γ is not a plane curve,**
then the minimization problem

$$\kappa_{2,\gamma,K} := \inf \left\{ \|B\|_{L^2(V)}^2 : (V, B, \Gamma) \in \widetilde{CV}_\gamma(\mathbb{R}^3), \right. \\ \left. \mathbb{M}(V) = 1, \text{ spt}\|V\| \subset K \right\} > 0.$$

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Proof. Assume by contradiction $\kappa_{2,\gamma,K} = 0$, consider energy minimizing sequence $\{(V_j, B_j, \Gamma_j)\}_{j \in \mathbb{N}}$. By **Compactness**, there exists $(V, B, \Gamma) \in \widetilde{CV}_\gamma(\mathbb{R}^3)$, to which the sequence subsequentially converges to, with

$$\text{spt}\|V_j\| \subset K, \quad \mathbb{M}(V_j) = 1 \quad \forall j, \quad \|B\|_{L^2(V)}^2 \leq \liminf \|B_j\|_{L^2(V_j)}^2 = 0.$$

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Thus $\text{spt}\|V\| \subset K$, $\|M\|(V) = 1$, contradicts **Constancy Lemma**. $\square$

Sketch of proof

Proof of Theorem: Given any immersion as required, say Σ .

Let (V, B, Γ) be the induced oriented varifold of $(\Sigma, A, \partial\Sigma)$.

- Then $(\frac{1}{|\Sigma|} V, B, \frac{1}{|\Sigma|} \Gamma) \in \widetilde{CV}_\gamma(\mathbb{R}^3)$.

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Following for the same reason as follows: an unoriented varifold V is said to have generalized mean curvature $\vec{H}$ if for any $X \in C_c^1(\mathbb{R}^3)$:

$$\delta V(X) = - \int \langle \vec{H}, X \rangle d\|V\|.$$

Then for any $\lambda > 0$,

$$\delta(\lambda V)(X) = - \int \langle \vec{H}, X \rangle d(\|\lambda V\|),$$

i.e., λV also has generalized mean curvature $\vec{H}$, with $\|\vec{H}\|_{L^2(\lambda V)}^2 = \int |\vec{H}|^2 d(\|\lambda V\|) = \lambda \|\vec{H}\|_{L^2(V)}^2$.

Sketch of proof

Proof continued: As above,

$$\|B\|_{L^2(\frac{1}{|\Sigma|}V)}^2 = |\Sigma|^{-1} \|A\|_{L^2(\Sigma)}^2.$$

Clearly,

$$\mathbb{M}\left(\frac{1}{|\Sigma|}V\right) = 1, \quad \text{spt}\left\|\frac{1}{|\Sigma|}V\right\| \subset K.$$

Applying [Proposition](#) (non-trivial minimizer),

$$|\Sigma|^{-1} \|A\|_{L^2(\Sigma)}^2 = \|B\|_{L^2(\frac{1}{|\Sigma|}V)}^2 \geq \kappa_{2,\gamma,K} > 0.$$

Thus

$$|\Sigma| \leq \frac{1}{\kappa_{2,\gamma,K}} \|A\|_{L^2(\Sigma)}^2 =: C \|A\|_{L^2(\Sigma)}^2.$$



(un)oriented varifolds

- **Unoriented varifolds:** Radon measures on the trivial unoriented Grassmannian bundle $\mathbb{R}^3 \times G(2, 3)$.
 - $P \in G(2, 3)$ is canonically identified as the associated orthogonal projection matrix, and $G(2, 3)$ is canonically embedded into $\mathbb{R}^{3 \times 3}$.

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 - $P \in G(2, 3)$ is canonically identified as the associated orthogonal projection matrix, and $G(2, 3)$ is canonically embedded into $\mathbb{R}^{3 \times 3}$.
- **Oriented varifolds:** Radon measures on the trivial oriented Grassmannian bundle $\mathbb{R}^3 \times \tilde{G}(2, 3)$.
 - $\tilde{G}(2, 3)$ can be canonically identified as $\mathbb{S}^2 \subset \mathbb{R}^3$.

Weak surface with prescribed boundary

How do we define weak surfaces with prescribed boundary?

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- Divergence theorem on smooth surfaces Σ : for any $X \in C_c^1(\mathbb{R}^3; \mathbb{R}^3)$,

$$\begin{aligned} & \int_{\Sigma} \operatorname{div}_{\Sigma} X(x) d\mathcal{H}^n(x) \\ &= \int_{\Sigma} \langle X(x), \vec{H}(x) \rangle d\mathcal{H}^n - \int_{\partial\Sigma} \langle X(x), \eta(x) \rangle d\mathcal{H}^{n-1}(x), \end{aligned}$$

where η is the **inwards pointing** unit co-normal.

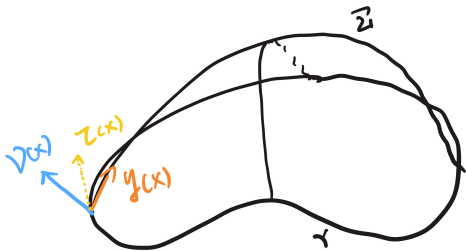
Remark. To formulate this integral formula in the measure-theoretic sense, we use **oriented varifolds**.

Prescribed boundary

- *Prescribed boundary condition:* Let τ be the tangent vector field on γ (C^2 , closed, embedded curve in $\mathbb{R}^3$).

Observation: if a smooth surface has boundary point $x \in \gamma$, then its normal $\nu(x) \perp \tau(x)$; inwards co-normal $\underbrace{\eta(x)}_{\perp \nu(x)} \perp \tau(x)$,

satisfies $\eta(x) = \tau(x) \times \nu(x)$;



Prescribed boundary

- Define at each $x \in \gamma$ the set $\tilde{G}_{2,\gamma}(x) := \{v \in \mathbb{S}^2 : v \perp \tau(x)\}$, which **provides all possible choices of normal**.
- Define the *generalized boundary bundle* $\tilde{G}_{2,\gamma}$ to be the collection of all such sets. Clearly $\tilde{G}_{2,\gamma} \subset \underbrace{\gamma \times \tilde{G}(2,3)}_{(x,v)}$.
- **Generalized boundary:** Radon measures on $\tilde{G}_{2,\gamma}$.
e.g., $(\mathcal{H}^1 \llcorner \gamma) \otimes \delta_{\nu(x)}$ is the boundary of a smooth surface Σ with $\partial \Sigma = \gamma$.
- **Generalized co-normal:** $\eta(x, v) := \tau(x) \times v$, a C^1 vector field on $\tilde{G}_{2,\gamma}$.

Prescribed boundary

- The boundary term can be written as (using disintegration)

$$\begin{aligned}
 & - \int_{\tilde{G}_{2,\gamma}} \langle X(x), \eta(x, \nu) \rangle d\Gamma(x, \nu) \\
 &= - \int_{\gamma} \left\langle X(x), \underbrace{\int_{\tilde{G}_{2,\gamma}(x)} (\tau(x) \times \nu) d\delta_{\nu(x)}(\nu)}_{=\tau(x) \times \nu(x) = \eta(x)} \right\rangle d\mathcal{H}^n(x)
 \end{aligned}$$

(Recall that $\Gamma = (\mathcal{H}^1 \llcorner \gamma) \otimes \delta_{\nu(x)}$).

- Hutchinson's** idea: Testing the divergence formula with $X(x) = \varphi(x, \nu(x))$, where $\varphi \in C_c^1(\mathbb{R}^3 \times \mathbb{R}^3; \mathbb{R}^3)$, gives the curvature identity.
 - Extra curvatures (except for mean curvature), comes from $\operatorname{div}_{\Sigma} X(x) = \sum \langle D_{\xi_i}(\varphi(x, \nu(x))), \xi_i \rangle$.

$$\widetilde{CV}_\gamma(\mathbb{R}^3)$$

- We say $(V, B, \Gamma) \in \widetilde{CV}_\gamma(\mathbb{R}^3)$ if for any $\phi \in C_c^1(\mathbb{R}^3 \times \mathbb{R}^3; \mathbb{R}^3)$:

$$\begin{aligned} & \int \langle \nabla_x \phi(x, v), v^\perp \rangle \\ & + \langle \phi(x, v), \text{Tr}(B)(x, v) \rangle \\ & + \nabla_v^* \phi(x, v) \cdot B(x, v) \, dV(x, v) \\ & = - \int_{\widetilde{G}_{2,\gamma}} \langle \phi(x, v), \boldsymbol{\eta}(x, v) \rangle \, d\Gamma(x, v). \end{aligned}$$

Some remarks

- **Lemma** (Boundary masses estimate):

$$\mathbb{M}(\Gamma) \leq C(\gamma) (\mathbb{M}(V) + \|B\|_{L^2(V)}).$$

Sketch of proof: Note that $\boldsymbol{\eta}(x, v) = \tau(x) \times v$ is C^1 on $\tilde{G}_{2,\gamma}$, by Kirszbraun's theorem $\exists$ Lipschitz extension $\tilde{\boldsymbol{\eta}}(x, v)$ to $\mathbb{R}^3 \times \mathbb{R}^3$.

Testing the curvature identity with $\phi(x, v) = l(x)\tilde{\boldsymbol{\eta}}(x, v)$, where $l(x)$ is a suitable cut-off function around γ ,

$$\mathbb{M}(\Gamma) = \int \underbrace{\langle \phi(x, v), \boldsymbol{\eta}_i(x, v) \rangle}_{\langle \boldsymbol{\eta}(x, v), \boldsymbol{\eta}(x, v) \rangle = 1} d\Gamma(x, v) \leq \text{curvature identity LHS.}$$

$\Rightarrow$ **Proposition** (Initial compactness).

Some remarks

- Let $\pi : \mathbb{R}^3 \times \widetilde{G}(2, 3) \rightarrow \mathbb{R}^3 \times G(2, 3)$ be the mapping from oriented Grassmannian to unoriented Grassmannian by forgetting the orientation.
If $(V, B, \Gamma) \in \widetilde{CV}_\gamma(\mathbb{R}^3)$ then $(\pi_* V, \pi_* \Gamma)$ agrees with the unoriented curvature with boundary by [Mantegazza 96'](#).

Thank you very much for your attention!